GinCol Lab · Aula

Student worksheet

Physical Education · 1st year of Bachillerato · Your body, your lab · Class 3 of 4 · With Isidora

Measuring yourself without getting hurt

Class video: https://youtu.be/b4JJHqm6jlY

The code leads to the video.

Name: _____ Date: _____

Before you start

1. Without looking at anything yet: if you've ever run the beep test, what do you think it measures? Your speed, your endurance, your determination not to give up? Write it down: it's just for you.

2. **At the end of the class.** Go back to what you wrote. What does it really measure, and how far can you trust your number?

While you watch the video

1. **The beep test.** At what speed does it start? How much faster does it get every minute? When does your test end, and which number is your score?

2. Why does it get faster every minute? What does that have to do with measuring yourself without getting hurt?

3. **What each test measures.** What's the difference between measuring yourself for health and measuring yourself to compete? Match each test in the ALPHA-Fitness battery with what it measures: heart and lungs, muscles, how you move, body composition.

4. How is the handgrip strength test done? And the standing long jump? And the 4 × 10 m shuttle run? For each one: how many attempts, and what is recorded.

5. Why were those tests chosen and not others? Give the three reasons.

6. **Does it measure well?** With the two kitchen scales: what is a reliable measurement? And a valid one? Can a measurement be valid without being reliable?

7. What is maximal oxygen uptake, and what unit is it measured in? How is it measured in a lab?

8. How did Léger get his equation? What two pieces of data does it need?

9. In Léger's equation, what happens to the estimate if you get further? And if, with the same stage, you're younger? Why does it make sense?

10. How far off can the equation be for one person? Why is it still useful for tracking yourself?

11. **Your number.** What is a percentile? Explain it with the line of a hundred.

12. Why is there one table for girls and another for boys? Does that mean that all boys get further than all girls?

13. In the 4 × 10 m, why is the 10th percentile among the fastest? And why isn't body composition ranked from best to worst?

14. What's the difference between reading your number in percentiles and reading it against the health threshold?

15. **Without getting hurt.** What is done before a maximal test? And during it? Why is it never done alone at home?

16. What is a fitness test for, first and foremost?

Reliable, valid or both?

Mark **R** if the measurement is reliable (it gives the same result when repeated) but not valid, **V+R** if it's both, and **N** if it isn't reliable.

1. A wristwatch that is always five minutes fast.

2. A new sewing tape measure, with which you measure the same table three times.

3. A thermometer that, with the same fever, shows 37.2, then 38.5 and then 36.9.

4. A kitchen scale that always shows 50 grams too little.

Practice

1. **Two percentiles.** A girl and a boy aged 16 (made up) both complete stage 5. Using the DAFIS table, roughly which percentile does each one fall in?

Stages at age 16	P10	P30	P50	P70	P90
Girls	1.83	2.98	3.89	4.90	6.62
Boys	3.08	5.46	7.16	8.77	10.86

1. **Two estimates.** Explain how your maximum heart rate formula (220 – age, or Tanaka's) and Léger's equation are alike.

2. **Léger's equation.** With a calculator, estimate the maximal oxygen uptake of a 17-year-old girl who stops during stage 6. $VO_2max \approx 31.025 + 3.238 \cdot V - 3.248 \cdot E + 0.1536 \cdot V \cdot E$, with V = 8 + 0.5 × stage (km/h) and E = age.

In your class, if you want to

These tests aren't done alone: they're done in class, with your teacher. If you do them, write down your numbers and come back to them in a few months, with the same rules. If you have an injury or a health problem, don't do them; if you feel dizzy or something isn't right, you stop.

Test	Today	In a few months
20 m shuttle run (completed stages)		
Handgrip strength (kg, each hand)		
Standing long jump (cm)		
4 × 10 m shuttle run (seconds)		

Your numbers are yours: nobody has to see them. To look yourself up in the tables for your age, FitBack (free; this link opens the Spanish version): https://www.fitbackeurope.eu/es-es/.

Keep thinking

Should your score in these tests count towards your Physical Education grade?

The data in the class come from the DAFIS study (Galicia, Iglesias-Soler et al., 2021, CC BY 4.0), from HELENA, from Ruiz et al. (2016) and from the studies cited in the teacher sheet.

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